

Print at 100% / Actual size, SINGLE-sided. Crease along the grey line, fold so both printed faces are outward, glue, then cut. This bar must measure exactly 80 mm.

[illegible]

NEW! SOLVE

Here's how? Do the walkthrough once at cubepath-six.vercel.app/learn. This one's the memory jog.

NOTATION
 R U' F L D B = turn that face clockwise, looking at it from outside the cube.
 r = counter-clockwise. 2 = half turn. Lowercase r f = that face plus the layer behind it. Turning together.
 y = spin the whole cube left. whole turns - Down. A whole-cube turn solves nothing.

WORDS
algorithm = a fixed sequence of words - trigger = a short algorithm you can use as a command

1 WHITE CROSS



White center DOWN: Bring the four white edges to it. Each edge's side color must match the center it touches. No algorithm – work it out.
 Can't see it? Build the four white edges on the YELLOW face first, each above its matching center, then turn each one down 180 degrees.

2 WHITE CORNERS

Find a white corner on turn U until it sits above the empty slot it belongs in, then match a picture below and repeat until it drops in.



white sticker RIGHT
R'U'R'

white sticker FRONT
L'U'L



white sticker UP turn the first one once to knock it out, then look again
R'U'R'

3 MIDDLE EDGES

Top edge with NO yellow: turn U until its front color matches the center under it. The arrow shows which slot it goes to.

goes RIGHT
U' R'U' y L'U'L



goes LEFT
U' L'U'L y' R'U'R'

Edge stick in the wrong slot? Run either one to kick it out, then place it.

■ **R'U'R'** sexy move
 ■ **R'U'R** Sune
 ■ **R'F'F'** 'sledgehammer
 gap = change grip

Legend:

- U**: Up (clockwise)
- U'**: Up (counter-clockwise)
- R**: Right (clockwise)
- R'**: Right (counter-clockwise)
- F**: Front (clockwise)
- F'**: Front (counter-clockwise)
- L**: Left (clockwise)
- L'**: Left (counter-clockwise)
- B**: Back (clockwise)
- B'**: Back (counter-clockwise)

Common Mistakes:

- 1. Don't twist the cube too much. Twist it only when necessary.
- 2. Don't twist the cube too much. Twist it only when necessary.
- 3. Don't twist the cube too much. Twist it only when necessary.
- 4. Don't twist the cube too much. Twist it only when necessary.
- 5. Don't twist the cube too much. Twist it only when necessary.
- 6. Don't twist the cube too much. Twist it only when necessary.
- 7. Don't twist the cube too much. Twist it only when necessary.
- 8. Don't twist the cube too much. Twist it only when necessary.
- 9. Don't twist the cube too much. Twist it only when necessary.
- 10. Don't twist the cube too much. Twist it only when necessary.
- 11. Don't twist the cube too much. Twist it only when necessary.

Sequence of Moves:

- Step 1:** Yellow cross. Moves: **F, U, F, U, F, U, F, U**
- Step 2:** Yellow corners. Moves: **R, U, R', U', R, U, R', U'**
- Step 3:** Yellow edges. Moves: **R, U, R', U', R, U, R', U'**
- Step 4:** White cross. Moves: **F, U, F, U, F, U, F, U**
- Step 5:** White corners. Moves: **R, U, R', U', R, U, R', U'**
- Step 6:** White edges. Moves: **R, U, R', U', R, U, R', U'**
- Step 7:** Middle layer. Moves: **R, U, R', U', R, U, R', U'**
- Step 8:** Last layer. Moves: **R, U, R', U', R, U, R', U'**
- Step 9:** Last layer. Moves: **R, U, R', U', R, U, R', U'**
- Step 10:** Last layer. Moves: **R, U, R', U', R, U, R', U'**
- Step 11:** Last layer. Moves: **R, U, R', U', R, U, R', U'**

ONE-LOOK PLI

ADJACENT CORNER SWAP

exactly one face shows headlights



• **T, L** headlights; pairs B-left F-left
RUR'U' R'F R2UR'U' RUR'F'



Ja L solid; pairs B-right F-left F-front
x R2FRF' RU2 R'F U2



Jb L solid; pairs B-left F-right B-back
RUR'F' RUR'U' R'F R2UR'F'



Aa L headlights; pairs F-right R-front
x L2D2 L'U'L D2 L'UL'



Ab L headlights; pairs B-right B-back
x L2D2 LUL' D2 LU'L



F L solid
F' L'F' RUR'U' R'F R2UR'U' RUR'UR'

ADJACENT CORNER SWAP

continued



Ra L headlights; pairs F-left
RUR'U' RUR DR'UR'D' R'U2R'



Rb U headlights; pairs B-front
R2F RURUR'R' F'R U2R'U2R



Ga L headlights; pairs F-right
R2UR'UR'U' RU'R2 U'D R'UR'D'



Gb L headlights; pairs B-back
R'UR' UD' R2UR'UR'U' RU'R2D



Ge L headlights; pairs B-right
R2UR'RU'R UR'R2 UD' R'UR'D



Gd L headlights; pairs R-front
RUR' U'D R2UR'RU' UR'UR2D'

One face shows headlights: two corners of that face match. Hold as drawn, run, then turn the top to finish.

■ RUR'U' sevy move
 ■ RURU' Sune
 ■ R'FRF' s'dedjeahmer
 gap = change grip

[illegible]

TWO-LOCK

OLL CROSS yellow edges



Line has left-to-right
FRUR'U'f'

Hook hook pointing front-right
FRUR'U'f'

OLL CORNERS

yellow face



Sune $\frac{1}{2}$ yellow; rest clockwise
RUR'U RU2R'

Anti-Sune $\frac{1}{2}$ yellow; rest counter-clockwise
RU2R' U'RUR'



Pi $\frac{1}{2}$ yellow; headlights left
FRUR'U'f' FRUR'U'f'

OLL CORNERS

continued



Headlights $\frac{1}{2}$ yellow; headlights at the back
R2R'U2 RU'R'U2R'



2x Headlights $\frac{1}{2}$ yellow; headlights both sides
RUR'U RU'R' URU2R'



Chameleon $\frac{1}{2}$ yellow next to each other
FRUR'U'f' FRF'



Bowtie $\frac{1}{2}$ yellow diagonal
FRURUR' FR

FALLBACK

The badge on each row is your fallback: that many Sunes, with a U turn between, also finishes the case. Machine-checked, three is the worst.
Top not all yellow and nothing matches? Run Sun and read again.

No yellow edge at all? Run the first one, then read again.

■ **FRUR'U'f'** ■ **RUR'U RU2R'** ■ **FRUR'U'f' FRUR'U'f'** ■ **FRUR'U'f' FRF'** ■ **FRURUR' FR** ■ **change sign**

[illegible]

ANNEX – BIG CUBES AND THE MAP ©2014

BIG CUBES 4x4 and 5x5

Reduce: build the 6 centers → pair the 12 edges → solve it as a 3x3. $Rw = 2$ layers wide, $3Rw = 3$ layers. $2R = 2nd$ LAYER only – never widen it.

last 2 edges

	R	U	R'	F	R'										
	2R2	U2	2R2	Uw2	2R2	Uw2		slice out, run it, slice back – both cubes							

4x4 PLL parity 1 edge pair flipped - half the time 2 edge pairs swapped - half the time

4x4 ULL parity	1 edge pair flipped - half the time														
Rw U2 x4	x Rw U2	Rw U2	Rw U2	Rw'	U2	Lw	U2	Rw'	U2	Rw U2	Rw' U2	Rw' U2	Rw' U2	Rw' U2	Rw' U2

4x4 edges parity 1 edge pair flipped - half the time **move** **one** move; 5x5 has no PLL parity

Rw U2 x4	x Rw U2	Rw U2	Rw U2	3Rw	U2	Lw	U2	Rw' U2	Rw U2	Rw' U2	Rw' U2	Rw' U2	Rw' U2	Rw' U2	Rw' U2
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5x5 parity: the last layer, before ULL, 3Rw – both parity algorithms, move corners

NOTATION

Each letter L that that face clockwise, seen from outside the cube – so from your seat, turn D and B look counter-clockwise – counter-clockwise = 2 half turn.

M = middle slice, turning the way L turns, $x \times 2 =$ the whole cube, turning the way R, U and F turn. Lowercase f f = that face plus the layer behind it, turning together.

BEGINNER SHORTCUT – use it until you know Sune

Yellow corner four turns at a time, turning only the top face between corners.

yellow faces RIGHT **D'R'D** **x2**

yellow faces FRONT **D'R'DR** **x2**

WORDS

algorithm = a fixed sequence of turns - trigger = a short algorithm you hands run as one unit - sexy move = the R U R' U' trigger - trigger = the R' F F' trigger, parity = a case a 3x3 cannot have - edge pair = 2 edges glued into one, on a 4x4 or 5x5

■ R U R' U' ■ R U R' U' ■ R F F' F' D = short algorithm, on a cube, rim